

Principles of communication systems

EET3202, CUNY City Tech, Fall 2023

Homework #03 Solutions

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Problem 1

Part 1

$$S(t) = m(t) \cos(\omega_c t + \theta)$$

$$e(t) = S(t) \cos(\omega_c t) = m(t) \cos(\omega_c t + \theta) \cos(\omega_c t)$$

use the following trigonometric identity

$$\cos \alpha \cos \beta = \frac{1}{2} (\cos(\alpha + \beta) + \cos(\alpha - \beta))$$

Therefore,

$$e(t) = \frac{1}{2} m(t) (\cos(2\omega_c t + \theta) + \cos(\theta))$$

Problem 1

Part 2

The low pass filter removes the term $\frac{1}{2}m(t)\cos(2\omega_c t + \theta)$ because it has a higher frequency than the filter cutoff ω_c .

Therefore, the final signal is

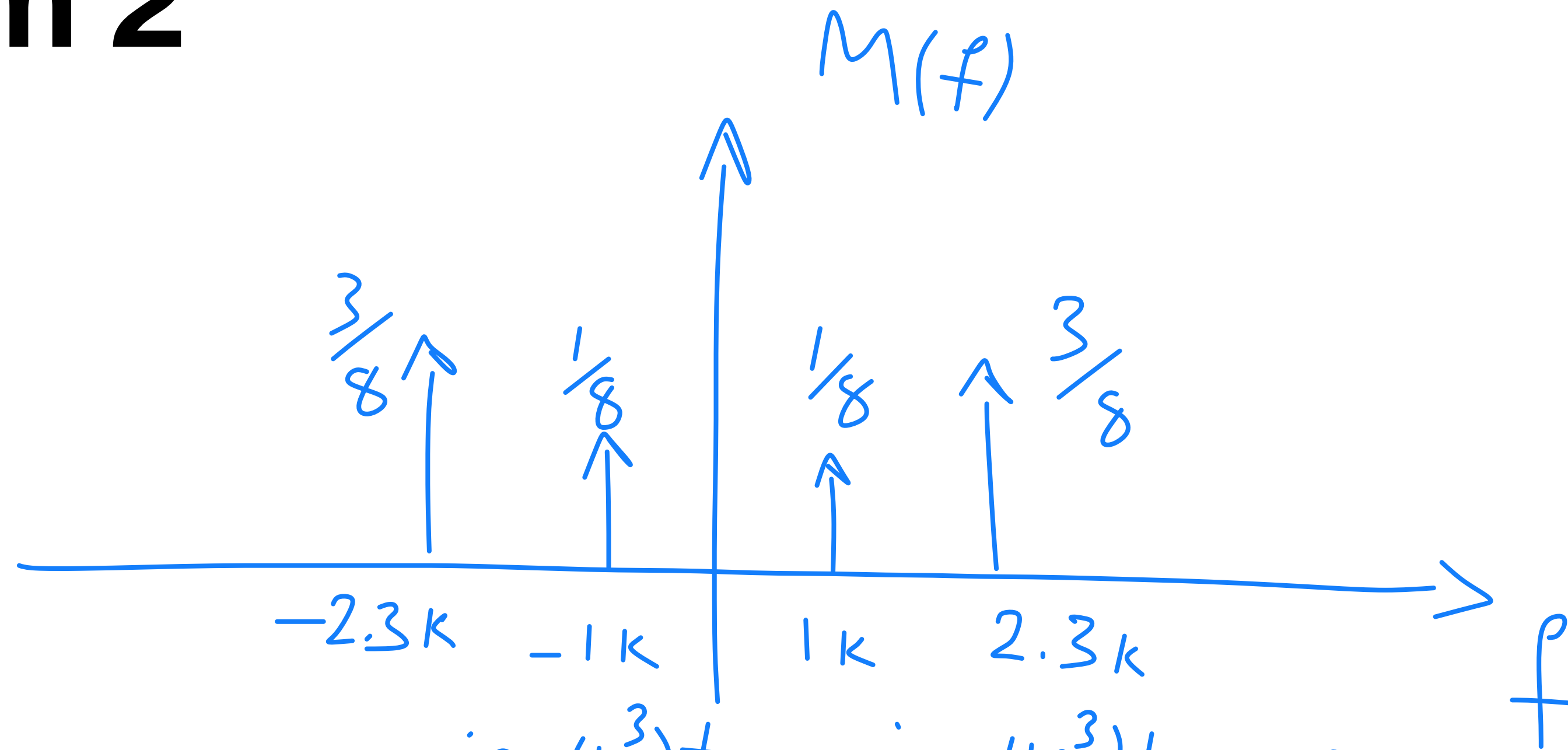
$$f(t) = \frac{1}{2} m(t) \cos(\theta)$$

Part 3 $\theta = \pi/2 \rightarrow \cos \theta = 0 \rightarrow f(t) = 0$

Part 4 The phase shift makes the coherent modulation a difficult task since it might remove the message entirely at the output of the low pass filter.

Problem 2

Part 1



$$\frac{1}{4} \cos(2\pi(10^3)t) = \frac{1}{8} (e^{j2\pi(10^3)t} + e^{-j2\pi(10^3)t}) \xleftrightarrow{\mathcal{F}} \frac{1}{8} (\delta(f-10^3) + \delta(f+10^3))$$

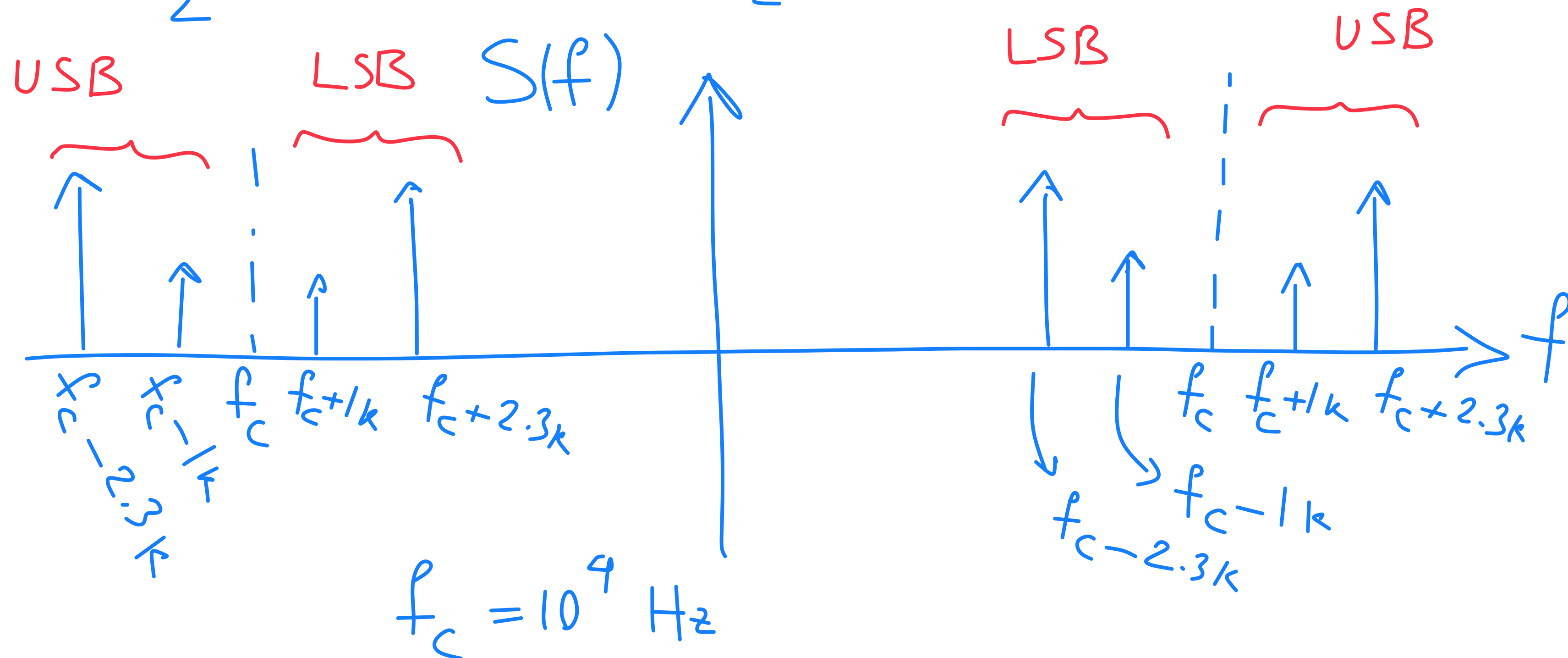
$$\frac{3}{4} \cos(2\pi(2.3 \times 10^3)t) \xleftrightarrow{\mathcal{F}} \frac{3}{8} (\delta(f-2.3 \times 10^3) + \delta(f+2.3 \times 10^3))$$

Problem 2

Part 2

$$s(t) = m(t) \cos(2\pi(10^4)t) = \frac{1}{2}m(t) \left(e^{j2\pi(10^4)t} + e^{-j2\pi(10^4)t} \right)$$

$$S(f) = \frac{1}{2}M(f-10^4) + \frac{1}{2}M(f+10^4)$$

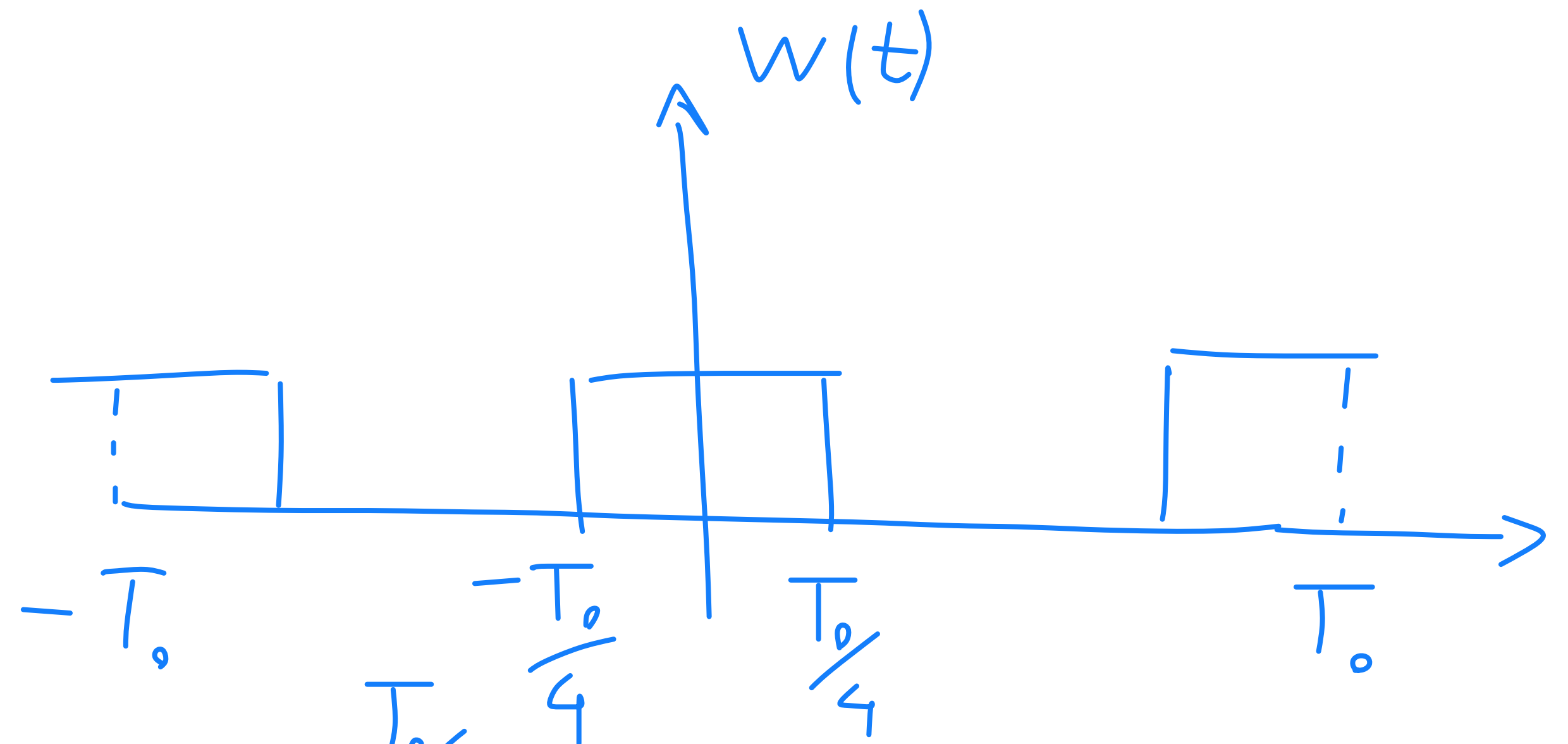


Problem 3

$$w(t) = \sum_{n=-\infty}^{\infty} C_n e^{j2\pi n f_0 t}$$

$$C_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} w(t) e^{-j2\pi n f_0 t} dt$$

$$= \frac{1}{T_0} \cdot \frac{1}{-j2\pi n f_0} e^{-j2\pi n f_0 t} \Big|_{-T_0/4}^{T_0/4} = \frac{1}{-j2\pi n f_0 T_0} (e^{-j2\pi n f_0 T_0/4} - e^{j2\pi n f_0 T_0/4})$$



Problem 3

By definition we know $f_0 = \frac{1}{T_0}$. Also we use $\frac{e^{j\theta} - e^{-j\theta}}{2j} = \sin(\theta)$

Therefore,

$$C_n = \frac{1}{-j2\pi n} \cdot \left(-2j \sin\left(\frac{2\pi n}{4}\right) \right)$$

$$C_n = \frac{\sin(\pi n/2)}{2\pi n/2} = \frac{1}{2} \operatorname{sinc}\left(\frac{n}{2}\right)$$

Finally, $C_0 = \frac{1}{2}$, $C_1 = C_{-1} = \frac{1}{\pi}$, $C_2 = C_{-2} = 0$

$$w(t) \approx \frac{1}{2} + \frac{1}{\pi} e^{j2\pi f_0 t} + \frac{1}{\pi} e^{-j2\pi f_0 t} = \frac{1}{2} + \frac{2}{\pi} \cos(2\pi f_0 t)$$